GA

ARISHA - 23122106

Date: M T W T F S S

Data Set

Set No.	Temperature	Humidity	Windy	Rainy
Sunny	Hot	High	Weak	NO
Sunny	mild	High	Strong	NO
Cloudy	Hot	High	Weak	Yes
Rainy	mild	High	Weak	Yes
Rainy	Cool	Normal	weak	Yes
Rainy	Cool	Normal	Strong	NO
Cloudy	mild	Normal	Strong	Yes
Sunny	Cool	Normal	Weak	NO

① ENTROPY CALCULATIONS OF PRIME ATTRIBUTES:

$$) - \frac{3}{8} \log_2 \frac{3}{8} = \frac{2}{8} \log_2 \frac{2}{8} - \frac{3}{8} \log_2 \frac{3}{8}$$

$H(S) = 1.56$

② INFO GAIN:

Info gain $(S, A) = H(S) - H(S/A)$

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*) Info gain of Out Look:

Outlook	Total	Yes	NO	entropy
Sunny	3	0	3	0.00
Cloudy	2	2	0	0.00
Rainy	3	2	1	0.91

The weighted entropy is.

$$H(S/A) = \frac{3}{8}(0.00) + \frac{2}{8}(0.00) + \frac{3}{8}(0.91)$$

$$H(S/A) \quad 0.341$$

info gain (S, Outlook) = 1.56 - 0.341

info gain = 1.219

*) INFO gain of Temperature

Temperature	Total	Yes	No	entropy
Hot	2	1	1	1
mild	3	2	1	0.91
Cool	3	1	2	0.91

The weighted entropy is.

$$H(S/Temp) = \frac{2}{8}(1) + \frac{3}{8}(0.91) + \frac{3}{8}(0.91)$$

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M	T	W	T	F	S	S

$H(S/Temp) = 0.935$

info gain $(S, Temp) = 1.56 - 0.935$

[info gain $= 0.625$]

*) INFO Gain of Humadity:

Humadity	Total	Yes	NO	entropy
High	4	2	2	1
Normal	4	2	2	1

The weighted entropy is

$H(S/Humadity) = \frac{4}{8}(1) + \frac{4}{8}(1)$

$H(S/Humadity) = 1$

info gain $(S, Humadity) = 1.56 - 1$

[info gain $= 0.56$]

*) INFO Gain of windy

windy	Total	Yes	NO	entropy
Strong	3	1	2	0.918
Weak	5	3	2	0.97

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The weighted entropy is.

$$H(S/windy) = \frac{3}{8}(0.918) + \frac{5}{8}(0.97)$$

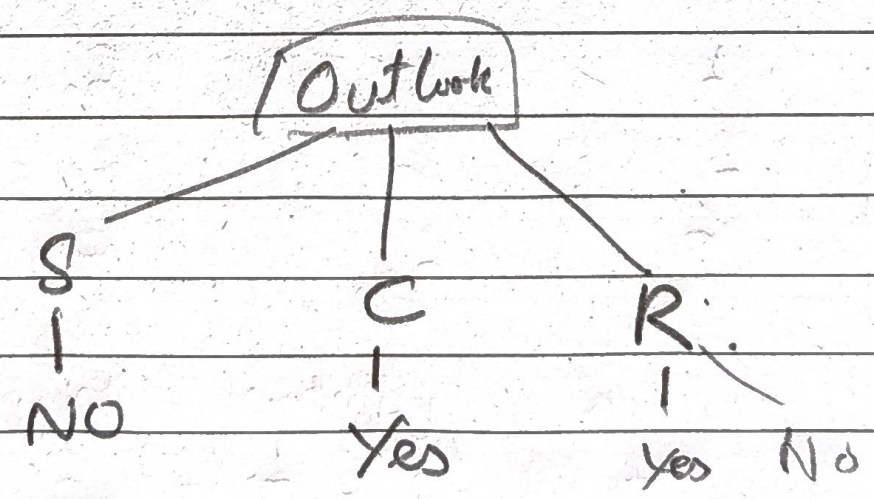
$$H(S/windy) = 0.95$$

info gain = 1.56 − 0.95

info gain = 0.61

(B) COMPARE INFO GAIN VALUE:

Attributes	Weighted entropy	Info gain
Outlook	0.341	1.219
Temperature	0.935	0.625
Humadity	1	0.56
Windy	0.95	0.61



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